Advanced SQL
NULL Functions
Exercise 2.1 - Practical SQL worksheet

Question 1: Show all Employees with their Salary. If Salary is NULL,

Table 1: Employees

Employee-id	Name	Department	Salary
1	Alice	HR	5000
2	Bob	IT	NULL
3	Charlie	NULL	7000
4	Dana	Finance	NULL

```
SELECT employee-id,
       name,
COALSE (salary, 0) AS salary-with-default
FROM Employees;
```

Output Table

Employee-id	Name	Salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Question 2: Show all employee name with their department. If department is Null show "Not Assigned".

Table 1: Employees

```
SELECT Employee_id,
       name,
COALESCE (department, 'Not Assigned') AS department_name
FROM Employees;
```

Output Table

Employee_id	Name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Question 3: Find orders with NULL customer_id using ISNULL()

Table 2: Orders

Order_id	Customer_id	delivery_date
101	201	2024-12-01
102	202	NULL
103	NULL	2024-12-03

```
SELECT  order_id,
        customer_id
FROM    orders
WHERE   customer_id IS NULL;
```

Output Table

Order_id	Customer_id
103	NULL

Question 4: Show all orders. If delivery-date is NULL showing 'Pending'

ORDER

Order_id	Customer_id	delivery-date
101	201	2024-12-01
102	202	NULL
103	NULL	2024-12-03

```
SELECT    order_id,
          Customer_id
COALSCE   (delivery-date, 'Pending') AS
          delivery-status
FROM      Orders;
```

Output Table

Order_id	Customer_id	delivery-status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Question 5: Show all students and their grades
Replace NULL with 0.

Table 3: Students

Student_id	Name	Grade
1	Ethan	85
2	Maya	NULL
3	Olivia	90

```
SELECT    student_id,
          name,
IFNULL (grade, 0) AS Final_grade
FROM      Students;
```

Output Table.

Student_id	Name	Final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Question 6: Count Students who haven't been graded.

Student

Student_id	Name	Grade
1	Ethan	85
2	Maya	NULL
3	Olivia	90

```
SELECT Count(*) As not_graded_count
FROM Student
WHERE grade IS NULL;
```

Output Table

not_grade_count
1

Question 7: Show product name, Price, and final price after discount (assume 0 if discount is NULL)

Table 4: Products

Product_id	Name	Price	discount
501	Keyboard	25	NULL
502	Mouse	15	5
503	Monitor	100	NULL

```
SELECT
        Product_id,
        name,
        (Price - COALSCE(discount, 0)) AS Final_price
FROM    Products;
```

Output Table

Product_id	name	Final_Price
501	keyboard	25
502	Mouse	10
503	Monitor	100

Question 8 Count how many customers have no Emails

Table 5: Customers

Customer_id	name	email
1	Linda	NULL
2	Joseph	joseph@mail.com
3	Nia	NULL

```
SELECT Count(*) AS missing_email_count
FROM Customers
WHERE email IS NULL;
```

Output Table

Missing_email_count
2

Question 9: Show all customers with email, if NULL, display "No email"

Customer_id	Name	email_display
1	Linda	No Email
2	Joseph	joseph@mail.com
3	Nia	No Email

Question 10: Show Payment details Method Replaced by "Unknown" if NULL

Table 6: Payments

Payment_id	Method	status
301	Credit	NULL
302	Paypal	Success
303	NULL	Failed

```
SELECT Payment_id
COALSE(method, 'unknown') AS Method_display, status
FROM Payments;
```

Output Table

Payment_id	Method_display	Status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed

Question 11: Show items and their quantity (0 if Null)

Table 7: Inventory

Item_id	Item_name	quantity
1	Pen	Null
2	Notebook	150
3	Eraser	Null

```
SELECT    Item_id,
          Item_name,
COALESCE (quantity, 0) AS quantity_checked
FROM Inventory;
```

Output Table

Item_id	Item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Question 12: Show Employee ID and the first available value among bonus or Commission.

Table 8: Employees_Extra

Emp-id	Bonus	Commission
1	NULL	300
2	100	NULL
3	NULL	NULL

```
SELECT
        Emp-id

COALESCE (bonus, Commission) AS first_Available_Reward

FROM Employees_extra;
```

Output Table

Empl_id	First_available_Reward
1	300
2	100
3	NULL

Question 13: Count classes that don't have a room assigned

Table: a classes

Class_id	Subject	room
11	Math	NULL
12	Science	Lab A
13	English	NULL

```
SELECT count(*) AS No_room_count
FROM classes
WHERE Room IS NULL;
```

output table

No_room_count
2

Question 14: Show attendance record with status. Replace NULL with "Not Marked".

Table 10: Attendance

Student_id	Date	Status
1	2025-04-01	NULL
2	2025-04-01	Present
3	2025-04-01	Absent

```
SELECT
    student_id,
    Date,
COALESCE(status, 'Not Marked') AS attendance_status
FROM Attendance;
```

Output Table

Student_id	Date	Attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Question 15: Show account ID, Account_type (OR 'unknown') and balance (OR 0).

Table 11: Bank_Accounts

Account_id	Account_type	Balance
A1	Savings	NULL
A2	Current	5000
A3	NULL	2000

```
SELECT
    account_id
COALESCE(account_type, 'unknown') AS type_display,
COALESCE(Balance, 0) AS balanced_checked
FROM Bank_Accounts;
```

Output Table

Account_id	type_display	balanced_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Question 16: Show all projects with a start date. If start_date is NULL, display 'TBD'.

Table 12: Projects

Project_id	Title	Start_date	End_date
1	Website Revamp	2025-01-10	NULL
2	Mobile App	NULL	2025-06-01
3	Data Migration	NULL	NULL

```
SELECT
    Project_id,
    title,
COALESCE(start_date, 'TBD') AS start_display
FROM Projects;
```

Output Table

Project_id	Title	Start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD

Question 17: Display Reviews showing Comment (OR 'No Comment') and rating (OR 0).

Table 13 Reviews

Review_id	Product_id	Comment	Rating
1	501	Great product	4
2	502	NULL	NULL
3	503	works fine	3

```
SELECT
        Review_id,
        Product_id,
COALESCE(Comment, 'No Comment') AS comment_Display,
COALESCE(Rating, 0) AS rating_display
FROM Reviews;
```

Output Table

Review_id	Product_id	Comment_display	Rating_Display
1	501	Great Product	4
2	502	No comment	0
3	503	works fine	3

Question 18: Show supplier contact number. Use COALESCE(phone, alt_phone, 'No Contact')

Table 14 Suppliers

Supplier_id	name	Phone	alt_phone
1	Global Goods	NULL	NULL
2	Best Supplies	987654321	NULL
3	ValueSource		NULL

```
SELECT
    Supplier_id,
    name,
COALESCE (phone, alt_phone, 'No Contact') AS Contact_number
FROM Suppliers
```

Output Table.

Supplier_id	name	Contact_number
1	Global Good	1234567789
2	Best Supplies	987654321
3	ValueSource	No Contact

Question 19: Show all users and their Preferences. Replace all Nulls with defaults:

Table 15: User_Settings

User_id	theme	language	timezone
1	NULL	English	NULL
2	Dark	NULL	UTC+1
3	NULL	NULL	NULL

```
SELECT
    user_id,
    COALSCE(theme, 'Light') AS theme_set,
    COALSCE(language, 'English') AS language_set,
    COALSCE(timezone, 'UTC') AS timezone_set

FROM User_Settings;
```

Output Table.

User_id	theme_set	languageSet	timezoneset
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

Question 20: Show maintenance log with:

Table 16: Maintenance

Record_id	Machine_id	Issue	Technician
1	M101	Overheating	NULL
2	M102	NULL	NULL
3	M103	Jammed	Alex

```
SELECT
    Record_id,
    Machine_id,
    COALESCE(issue, 'unknown Issue') AS Issue_log,
    COALESCE(technician, 'Not Assigned') AS technician_name

FROM Maintenance;
```

Output Table

Record_id	machine_id	Issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex